

In this project, I tried to build a playful driving simulator that takes the familiar idea of a car road trip and combines it with the mechanics of drum sets, because I want to add some differences to distinguish this and regular drive simulator by adding these mechanics to make it a fidget game. Instead of the car being controlled by a traditional steering wheel or WASD keys, I designed a system where four drums sit in front of the player. Each drum is mapped to a different key, and pressing it produces a unique drum sound while also directing the car. For example, one drum acts as the accelerator, another as the brake, and the others control turning. What I really liked about this concept is that it creates a cozy atmosphere by mixing the background music and drum sounds. The result feels less like a racing game and more like an interactive performance, where the scenery outside the window and the drum sounds together create a kind of road trip rhythm experience.

The process was far from smooth. One of the biggest challenges I faced was finding the right balance between realism and smooth driving. In reality, driving across sand is steep, bumpy, and often unpredictable, but when I tried to replicate that in Unity, the experience became frustrating and broke the cozy, playful atmosphere I wanted my game to have. Unity's physics system made it difficult to keep the car grounded without it feeling too heavy or chaotic. I experimented with Rigidbody settings, collision detection modes, and constraints, but what really helped was deciding to prioritize smoothness over realism. I locked the car's unnecessary rotations, carefully adjusted interpolation, and then added some camera shake to simulate the sense of bumps without making the gameplay unpleasant. To solve these problems, I relied on online tutorials in the youtube, and messed up with rigidbody physics.

If I reflect on my process, I think one thing I would do differently is to decide on the style of the game much earlier and make sure I implement it from the beginning. Based on today's classmates' review, I also see the importance of adding extra features that make the driving feel more complete, such as a horn sound, which feels like an essential element that both supports the rhythm aspect and makes the car feel more alive. Watching the demos from graduating students in class today also gave me perspective on how much more aspects I could add. Their projects reminded me that small details, like sideview mirrors or other subtle interface touches, contribute a lot to immersion. Going forward, I want to push myself not only to refine the core mechanic but also to enhance the surrounding features that make the experience richer and more believable.

Looking toward the future, I think this game concept has potential to grow. Right now it feels like a quick experiment, but with further development it could evolve into a rhythm-driving hybrid experience. Imagine traveling through diverse landscapes that deserts, grasslands, cities, while each environment changes the instrument set.